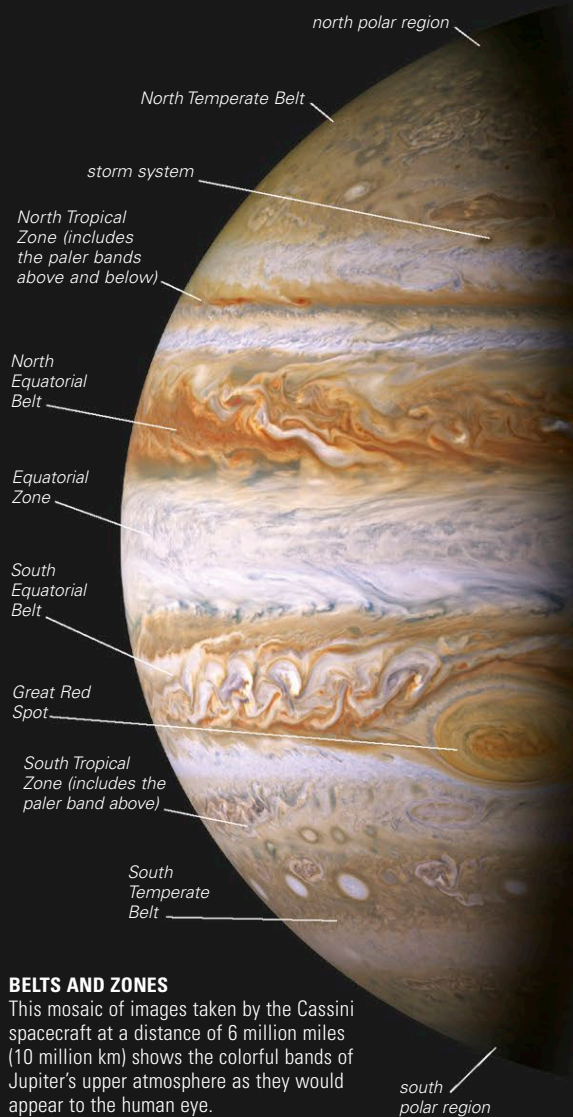


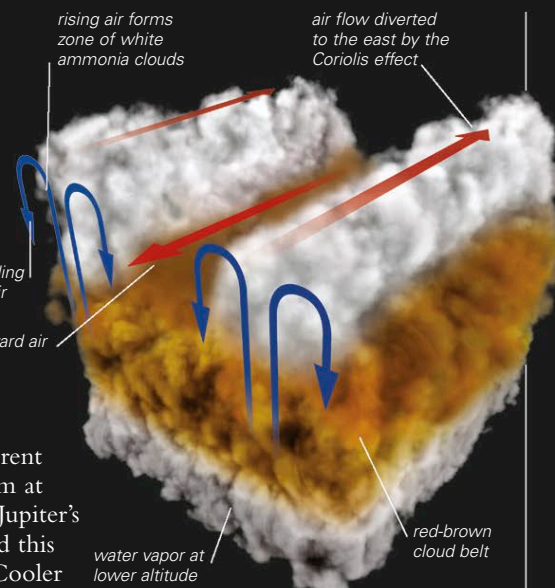


BELTS AND ZONES

This mosaic of images taken by the Cassini spacecraft at a distance of 6 million miles (10 million km) shows the colorful bands of Jupiter's upper atmosphere as they would appear to the human eye.

ATMOSPHERE

Jupiter's atmosphere is dominated by hydrogen, with helium being the next most common gas. The rest is made up of simple hydrogen compounds—such as methane, ammonia, and water—and more complex ones such as ethane, acetylene, and propane. It is these compounds that condense to form the different-colored clouds of the upper atmosphere and help give Jupiter its distinctive banded appearance. The temperature of the atmosphere increases toward the planet's interior. As gases condense at different temperatures, different types of clouds form at specific altitudes. All the while, the gas in Jupiter's equatorial region is heated by the Sun, and this rises and moves toward the polar regions. Cooler air flows from the polar regions at a lower altitude to take its place, creating in effect a large circulation cell. This hemisphere-wide circulation transfer would be straightforward if Jupiter were stationary. It is not—it rotates, and speedily at that, and a force known as the Coriolis effect (see p.126) deflects the north-south flow into an east-west flow. As a result, the large circulation cell is split into many smaller cells of rising and falling air. These are seen on Jupiter's surface as alternating bands of color. Jupiter's white bands of cool rising air are called zones. The red-brown bands of warmer falling air are known as belts.

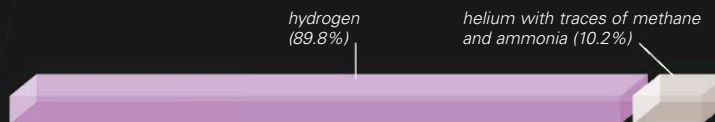


CLOUD FORMATION

Clouds of different compounds form at different altitudes in the atmosphere. Convection currents move the mixture of gas upward. Water is first to reach the altitude where it is cool enough to condense to form clouds. Higher up, where it is cooler, red-brown ammonium hydrosulfide clouds form, and highest of all, where it is coolest, are the white ammonia clouds.

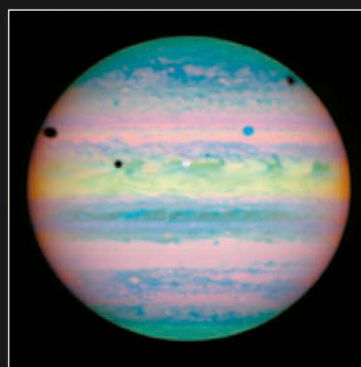
COMPOSITION OF ATMOSPHERE

Hydrogen dominates, but it is the trace compounds that color Jupiter's upper atmosphere.



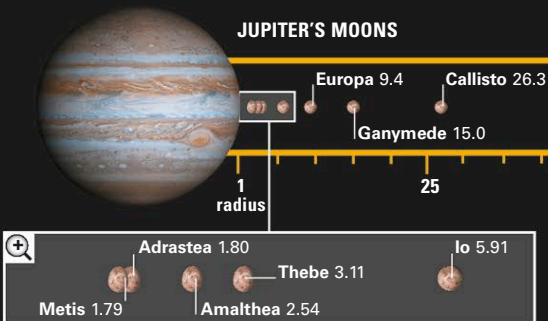
MOONS

Jupiter has over 79 known moons, over two-thirds of which have been discovered since January 2000. Only 50 of the moons have been given names, and several have yet to have their orbit confirmed. The recent discoveries are typically irregularly shaped rocky bodies a few miles across and are thought to be captured asteroids. By contrast, Jupiter's four largest moons are spherical bodies that were formed at the same time as Jupiter. Collectively known as the Galilean Moons (see p.182), they were the first moons to be discovered after Earth's Moon. As they orbit Jupiter, passing between it and the Sun, their shadows sweep across the planetary surface; from within the shadow, the Sun appears eclipsed. A triple eclipse happens just once or twice a decade.



TRIPLE ECLIPSE

Three shadows were cast onto Jupiter's surface on March 28, 2004, as its three largest moons passed between the planet and the Sun. Io is the white circle in the center, with its shadow to its left. Ganymede is the blue circle at upper right, and its shadow lies on Jupiter's left edge. Callisto's shadow is on the upper-right edge, but the moon itself is out of view, to the right of the planet.



Scale in radii of Jupiter
1 radius = 44,397 miles (71,492 km)



Moons are not to scale and increase in size for magnification purposes only